# Orthographic projection

National 5 · Drawing & spatial awareness · Worksheet

GRAPHICS WITH MR KELLY

Name _____ Class _____ Date _____

**By the end of this worksheet I can…**

- ☐ identify and arrange elevation, plan and end elevation in third-angle projection
- ☐ show hidden detail correctly in every view
- ☐ work out missing sizes from an orthographic drawing
- ☐ identify true lengths and true shapes

## I DO — Worked example

Read how a full-mark answer is built

*An L-shaped bracket is shown as a pictorial view. Four sets of orthographic views are shown, labelled A to D.*

**1 State what you would check to decide which set of views is correctly arranged in third-angle projection.** (2 marks)

**Points that earn marks** from the marking instructions: one ✓ per point

- ✓ The plan is drawn above the elevation
- ✓ The end elevation is on the side it is viewed from

## WE DO — Guided practice

Use the starters and word bank to help you

*A shelf bracket has an overall length of 200. From the left it has a 62 flange, then a slot, then a 72 end section.*

**2 Calculate the length of the slot.** (1 mark)

| Sentence starters | Word bank |
|---|---|
| One … is …, which means … | |
| Another … is …, so … | |

Tip: 1 marks = 1 separate, developed points.

## YOU DO — On your own

No notes. Check your answers afterwards.

**3 Quick check: tick one box for each.**

**a)** In third-angle projection, the plan is drawn…
☐ Below the elevation ☐ Above the elevation ☐ To the left of the elevation ☐ On a separate sheet

**b)** Hidden detail is shown with…
☐ Thin dashed lines ☐ Thick continuous lines ☐ Chain lines ☐ Wavy lines

**c)** To find the true shape of a sloping cut you look…
☐ From the front ☐ From above ☐ At 90° to the cut surface ☐ From below

**d)** The third-angle symbol shows a cone frustum with its circles…
☐ On the left ☐ On the right ☐ Above ☐ Below

**4 Key terms: write the letter of the matching description next to each term.**

| | | |
|---|---|---|
| 1. **Third-angle projection: where is the plan?** | ____ | **A** Above the elevation. |
| 2. **Elevation** | ____ | **B** The main front view of an object. |
| 3. **End elevation** | ____ | **C** Views lined up so sizes project between them. |
| 4. **Related views** | ____ | **D** The side view, drawn on the side it is viewed from in third angle. |

**5 Explain what is meant by ‘Hidden detail’.** (2 marks)

**6 Explain what is meant by ‘True shape’.** (2 marks)

**Check and reflect**
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?